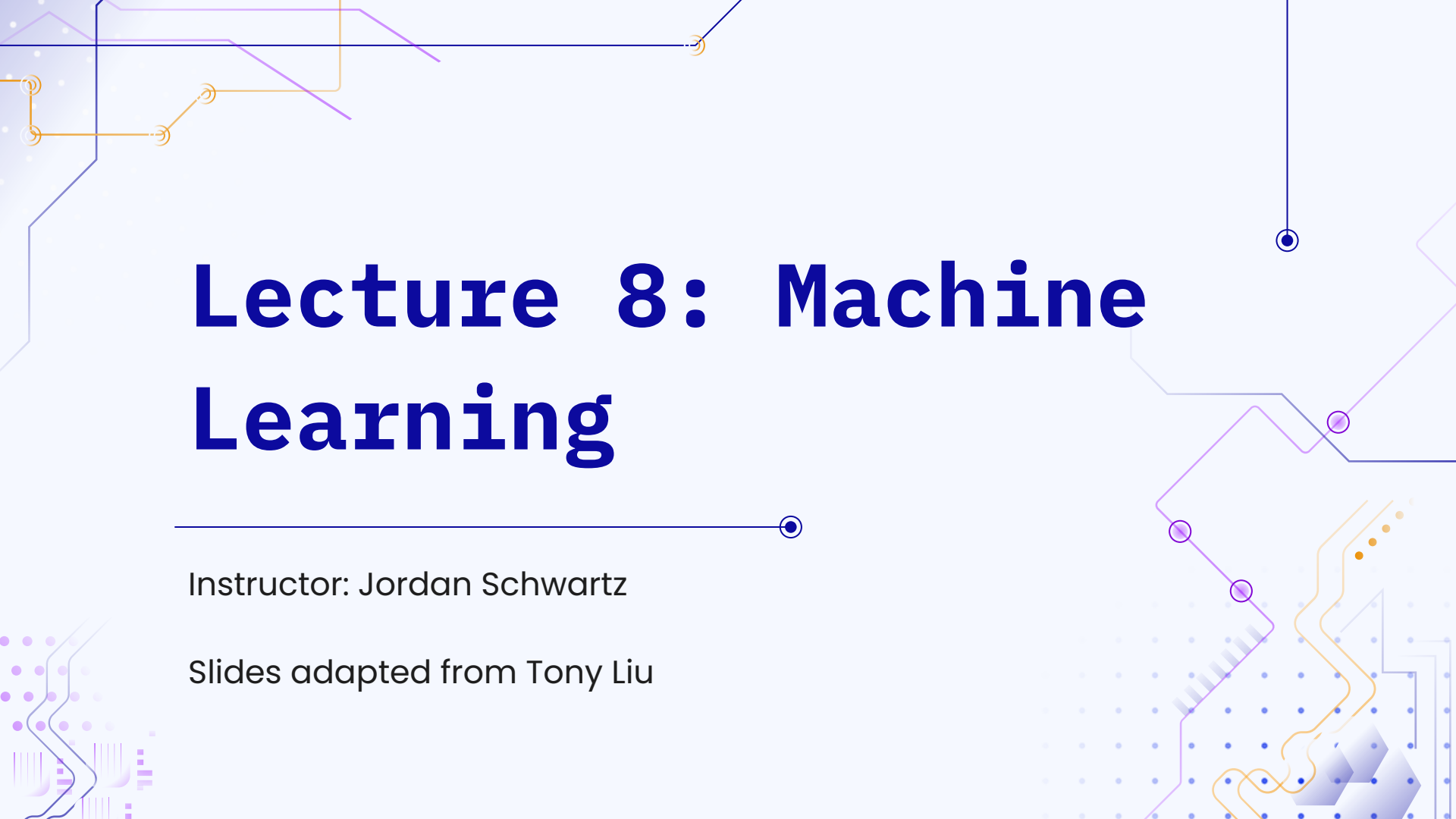




Lecture 8: Machine Learning

Instructor: Jordan Schwartz

Slides adapted from Tony Liu

Logistics

- Final project spec released. The first due date is one week from today. You will need to sign up for a check in next week. I will post these links after class today.
 - <https://calendly.com/alevaku-seas/cis-1902-project-check-ins>
 - <https://calendar.app.google/dFiXPf9EMixbCxrX8>

Week 6-7

- Jupyter/Colab
- Numpy
- Pandas
- Matplotlib

Week 8

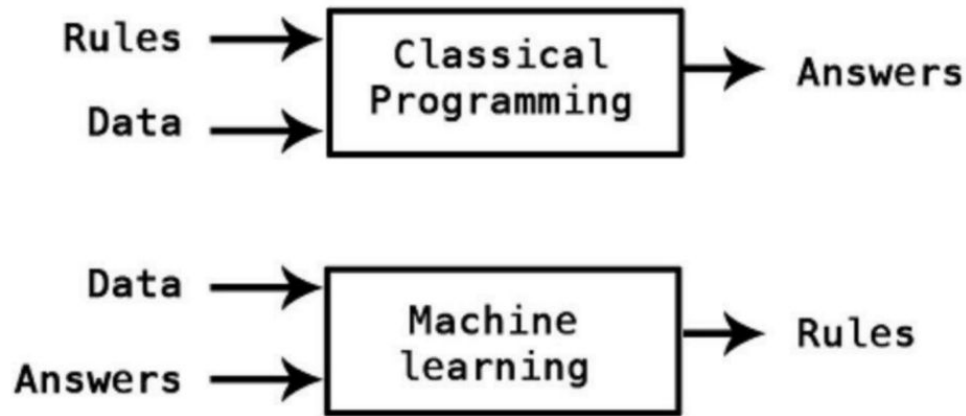
- What is machine learning?
- Different types of machine learning
- Defining a model
- Disclaimer about math
- Break it into:
 - supervised learning
 - Classification
 - Regression
 - The full machine learning process



Machine Learning

What is machine learning?

machine learning is **prediction** using data, by finding **patterns** in the data



Source: Francois Chollet

Different Types of Machine Learning

supervised learning (regression, classification)

given **data** and
answers, learn a
rule

Labeled data, e.g.
past exams with
solutions

unsupervised learning (clustering)

only given **data**, learn
a **rule** for the **patterns**
you see in the data

Unlabeled data, e.g.
just past exams

reinforcement learning

try out your **rule** on a
piece of data, and get
feedback on how
good your rule was

Getting feedback as
you take the exam

Defining a Model

- **Problem Type:** What is the goal of the output? Classification or value prediction?
- **Feature selection:** What are some insights about that data that are useful?
- **Model selection:** Given what I know above, what is the best class of model to pick for my problem?

The Full Machine Learning Process

1. data collection and cleaning
2. **data preprocessing**
3. **model training**
4. **model evaluation**
5. model deployment -> stay tuned for web dev section!

DISCLAIMER!

These next two lectures may get a bit involved mathematically. Basic linear algebra and probability are assumed knowledge.

I will not be testing you on any of this, it is purely for your own understanding!



Supervised Learning

Supervised Learning

- 2 types:
 - Classification
 - Regression
- Given a set of data points, each with features/descriptors/attributes, collect them into a $n \times d$ matrix X
 - example: we have n houses and 3 features: square footage, number of bedrooms, and year built
- we've described our data, now we need answers

Classification

- answers are discrete labels (can be more than 2),
 - such as whether the house will appreciate or depreciate in value
- we code the class labels as integers
 - 0 = "depreciate"
 - 1 = "appreciate"
 - y is a vector of shape containing 0s and 1s: $y \in \{0, 1\}^n$

Regression

- answers are continuous values, such as the sale price of the house
- y_n is a vector of shape containing real outputs: $y \in \mathbb{R}^n$

Which type of ML is best for each scenario?

M1

Identifying the Mood of a Rubber Duck

Imagine a computer vision system that classifies the facial expression printed on rubber ducks in a toy factory.

M2

Predicting the Ideal Marshmallow Toast Time

Suppose a smart campfire device tries to predict how long a marshmallow should stay over the fire before reaching the perfect toast level.

M3

Training a Robot to Catch Falling Pizza

A robot stands under a conveyor belt where pizzas randomly fall.

A: Regression
B: Classification
C: Clustering
D: Reinforcement Learning

Goal: turn new data points into predictions

y: classification	y: regression	x_1 : sq ft	x_2 : # bedrooms	x_3 : yr built
appreciate	\$500,000	2500	3	1999
depreciate	\$1,000,000	4000	5	1950
depreciate	\$250,000	1000	2	1980
appreciate	\$300,000	900	2	2010
?	?	2000	3	2015
...	...	...	...	...

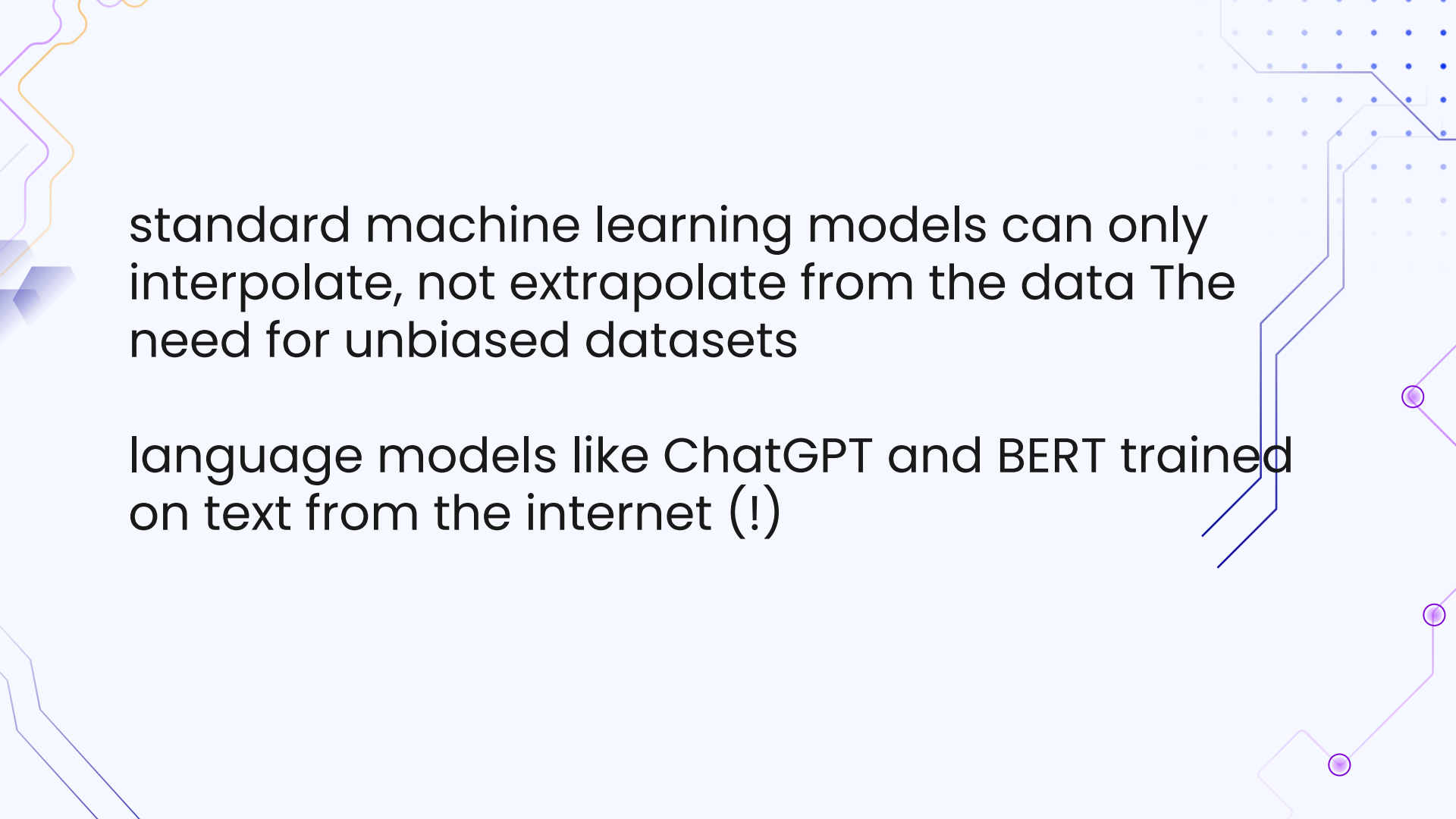
goal: learn a function f that takes *previously unseen* examples X_{new} and makes predictions $\hat{y} = f(X_{new})$

How do we learn f ?

- We need to find a "good" f function, somehow
 - there is no "correct" answer, so finding f is often framed as a optimization problem
 - leads to a wide variety of machine learning algorithms
- today, two models:
 - regression: ordinary least squares (OLS)
 - classification: k-nearest neighbors

Let's see an example!

<https://colab.research.google.com/drive/1eGTmENPqJlZVKA0FkoqYzSYQ7jQo3Wee>



standard machine learning models can only
interpolate, not extrapolate from the data The
need for unbiased datasets

language models like ChatGPT and BERT trained
on text from the internet (!)

http
458



Thanks !

Do you have any questions?

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